26 A coil produces a magnetic flux of 0.85 Wb. A second coil of 1500 turns is wound on the same core, such that all the magnetic flux from the first coil links the second coil.

What is the e.m.f. induced across the terminals of the second coil when the flux produced by the first coil is reduced uniformly to zero in 0.19 s?

A 0 V **B** 4.47 V **C** 335 V **D** 6710 V

Ans: D

The induced e.m.f. in the second coil is given by: $\varepsilon = N\frac{\Delta\Phi}{\Delta t}$

where N is the number of turns, $\Delta\Phi$ is the change in flux, and Δt is the time taken. $\varepsilon = 1500 \times \frac{0.85}{0.19} = 6711\ \text{V} \approx 6710\ \text{V}$